Subhrajit **Prusty**

LEAD BACKEND ENGINEER

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Summary

Backend & infrastructure engineer with 5+ years shipping and operating high-availability systems in Go and Python on AWS and Kubernetes (EKS). Build distributed services from 0 to 1 and operate them at scale: tens of millions of calls a month and ~99.5% uptime, backed by CI/CD, autoscaling, and observability that keep them reliable and cost-efficient. Deep hands-on experience with real-time voice AI, LLM-powered systems, and the GPU inference serving behind them.

Experience

Bolna

Bengaluru, India

LEAD BACKEND ENGINEER

January, 2026 - Present

- Own the platform infrastructure behind a voice-AI product handling ~34M calls a month across ~2,400 tenants: multi-region Kubernetes (API, WebSocket, and GPU-inference clusters) scaled ~5x in six months; own capacity planning, cluster and custom-metric autoscaling on application signals, ALB routing for latency-sensitive WebSocket and SIP traffic, and the cloud bill; cut the AWS run rate ~40% through cost optimization and cross-cloud GPU inference routing
- Own the security posture: an ISO-aligned, least-privilege, federated-identity model across every workload, host, CI pipeline, and developer (per-workload IRSA, per-instance EC2 instance profiles, exact-subject OIDC trust for CI, MFA-gated SSM-only host access with SSH closed on staging), centralized secrets via External Secrets Operator scoped per cluster, policy-simulation-verified IAM and RBAC, internal endpoints shut at the load balancer, and a fleet-wide credential-rotation procedure with per-workload verification
- Own the self-hosted speech-recognition (ASR) inference platform on NVIDIA L4 GPUs on the live call path: model-bundle lifecycle and version pinning, licensing and model storage, and autoscaling to real-time stream capacity on two clouds; built a GKE failover cluster from the EKS production values to survive AWS L4 stockouts, designed the weighted per-call ASR router across self-hosted AWS, self-hosted GCP, and Deepgram's managed endpoint (eligibility guards, served-endpoint cost attribution) that enabled production to move from 100% AWS to a three-endpoint split (the largest single line of that ~40% cut), and specified connect-time failover after showing a capacity-spillover router could not survive endpoint-side flaps
- Defined the new-service paved road: per-service databases with rw/ro roles, IRSA, immutable arm64 images, shared reusable CI workflows, OpenTelemetry, internal-domain and cert conventions (Argo CD as the GitOps end state, IRSA kept over EKS Pod Identity), and used it to take the first two internal services (an LLM evaluation harness and a cross-call agent-memory service) to production; added Azure OpenAI as a post-call extraction provider; isolated tenant-level LLM provider failures (non-retryable quota errors, per-tenant concurrency caps) so one customer's provider limits cannot degrade the shared worker fleet
- Took Bolna's browser-calling (WebRTC) media plane, FreeSWITCH behind a Kamailio SIP/WSS edge bridging browser audio into the voice pipeline, from hand-built boxes to an autoscaling fleet (golden AMI, drain hooks, CI deploys over SSM, Grafana and Datadog alerting) and validated it with a ~100-concurrent-call production load test with zero media failures
- Own the on-call and incident-management process: severity-classified incident reviews with runbooks linked from the alerts they serve; a measured MTTR/MTTD baseline across the full incident history that informed the contractual Sev-1/2/3 resolution-time SLAs and showed detection, not response, dominated time-to-resolve; the detection-first fixes that followed (volume-gated ratio alerts back-tested over 14 days, provider-attributed failure metrics so vendor outages page separately, memory monitors that fire ~9 minutes earlier); and command of multi-hour cross-stack incidents from detection through postmortem and follow-ups
- Own the telemetry data pipeline: shipped OpenTelemetry metrics and traces from 17+ workloads across four clusters into ClickHouse (ClickStack) under a 30-day retention policy, and made OpenTelemetry server-span instrumentation coexist with the Datadog tracer fleet-wide; moved backfill and analytical queries off primary Postgres onto the CDC-mirrored store and retired redundant CDC replication from the production database; applied lifecycle governance to unbounded registries and asset stores, extended PII-masking enforcement to every worker path, and put review guardrails (mandatory dry runs, undo capture, code ownership) on data-migration scripts
- Trace infra-level symptoms (OOM-killed pods, stalled event loops, connection-pool exhaustion, cost regressions) to their root cause in application code and fix both layers: cut a queue worker's per-pod memory ~95%, hardened the WebSocket fleet's health signal against event-loop stalls, and cut LLM client p99 latency ~75% under +54% traffic
- Re-architected post-call processing (CDR, settlement, disposition analytics) into a restart-safe, self-reconciling pipeline (atomic claim-and-checkpoint, reconciliation jobs that recover interrupted calls, durable queued admin retriggers, and bounded provider retries), giving idempotent settlement (no double billing) and safe horizontal scale across concurrent, multi-pod execution
- Built **kubedump**, an open-source Go CLI that snapshots live cluster state into clean, Git-trackable manifests on a daily CI schedule, the backbone for config drift-tracking and audit trails across all clusters

skit.ai *Bengaluru, India*

SOFTWARE ENGINEER - PRODUCT *November 2022 - August 2025*

- Engineered an LLM-powered voicebot in Go, processing millions of calls for 40+ enterprise clients; rebuilt it on the open-source Pipecat-AI framework for modularity and third-party integration.
- Designed and implemented a bidirectional audio-streaming telephony stack with gRPC for real-time interactions.
- Configured and maintained in-house + OpenSIPS infrastructure for call routing with telephony vendors like TCN and Twilio.
- Designed and implemented a flow-configuration validation service, completely eliminating config-related incidents, and optimized flow-configuration loading to cut memory usage by 80%.
- Owned the core voicebot and telephony services in production and was the de facto on-call escalation point for every voice-call incident, maintaining 99.5% uptime with Grafana, Loki, OpenTelemetry, and Prometheus.

skit.ai *Bengaluru, India*

SOFTWARE ENGINEER - SOLUTIONS *December 2020 - October 2022*

- Owned and maintained a critical internal repository, establishing code review processes and authoring contribution guidelines; significantly improving code quality and reducing technical debt.
- Developed 3rd-party API integrations, including PCI-DSS compliant Payment APIs, improving voice bot's debt collection capabilities
- Enhanced build pipelines, reducing deployment time and improving CI/CD efficiency.
- Maintained, customized and enhanced internal ML models used for real time conversational voice bots.
- Designed and implemented an automated daily reports system that simplified accessing and transforming on-prem data

racetrack.ai (Internship) *Bengaluru, India*

RESEARCH AND DEVELOPMENT TEAM *August 2019 - March 2020*

- Using data collected by automated web-scraping, published an abstract on Personalized Hybrid-Recommender Systems using social network analysis.
- Using realtime image processing, developed a minor claims automation service for car accident processing, improving claim resolution efficiency.

ITC Infotech, Innoruption Lab (Internship) *Bengaluru, India*

RESEARCH AND DEVELOPMENT TEAM *May 2019 - July 2019*

- Using image processing and automation, developed a metal sheet counting algorithm.
- Using facial detection and recognition, built an ID-card detail extraction system, optimizing concierge check-in process.

Education

Institute of Technical Education and Research (ITER) *Bhubaneswar, Odisha, India*

B.TECH IN COMPUTER SCIENCE AND INFORMATION TECHNOLOGY *2016 - 2020*

- CGPA : 8.81

Ispat English Medium School (IEMS) *Rourkela, Odisha, India*

INTERMEDIATE/+2 FROM ISC *2014 - 2016*

- Percentage : 84.5

Ispat English Medium School *Rourkela, Odisha, India*

MATRICULATION FROM ICSE *2014*

- Percentage : 94.17

Skills

Languages	Python, Go, Bash, SQL
Infra & Cloud	Kubernetes (EKS, GKE), Helm, Docker, External Secrets Operator, AWS (EC2/ASG, IAM/IRSA, OIDC, Secrets Manager, S3, ALB, Lambda, EFS, Budgets), GCP (GKE, Compute Engine, Filestore), NVIDIA GPU nodes, golden AMIs, GitHub Actions
ML Infrastructure	Self-hosted ASR inference (Deepgram Nova-3/Flux on L4 GPUs), GPU autoscaling, multi-cloud model serving (EKS/GKE/Modal), inference traffic routing and cost attribution, LLM provider routing (OpenAI, Azure OpenAI, Anthropic)
Observability	Datadog, Grafana, Loki, Prometheus, OpenTelemetry, Sentry, ClickHouse, ClickStack, Metabase
Backend & Messaging	FastAPI, Flask, gRPC, WebSockets, RabbitMQ, Redis, PostgreSQL, SIP/OpenSIPS, FreeSWITCH, Kamailio, Asterisk, WebRTC